MineArena 1.0 — Final Architecture V4.1

Manifest-Driven, Context-Adaptive Physics-Informed Cognitive Patient Digital Twin

Problem Statement: Personalized Post-Discharge Deterioration Monitor

Objective: Build an intelligent, personalized patient monitoring and recovery platform that understands each patient's clinical context, continuously analyzes physiological behavior, reconstructs a virtual physiological state using a textual manifest, predicts future physiological changes using PINN-ODEs, verifies its intelligence through a trustworthy layer, and provides evidence-based recommendations to clinicians and patients.

The architecture has **seven specialized experts, two knowledge bases, a manifest-driven PINN-ODE digital twin, trustworthy intelligence, and a human-supervised decision system.**

---

1. Final complete architecture

Complete System Architecture

Context-adaptive · Manifest-driven · Physics-informed · Trustworthy

LAYER 1 — DATA SOURCES

MATLAB Virtual Wearable

HR, RR, SpO_2, temperature, physiological scenarios

BIDMC Dataset

Real HR, RR and SpO_2 recordings

Synthea Clinical Registry

Patient records and clinical history

Patient Recovery Inputs

Activities, symptoms, medication reports and tasks

↓

LAYER 2 — SECURE COMBINED GATEWAY

Patient identity verification

Authentication & authorization

FHIR validation

Integrity & replay protection

Signal quality monitoring

Patient-aware load balancing

Adaptive 5/10/15s windows

Missingness & artifact handling

↓

HOT PATH

FastAPI

Real-time processing

Immediate safety checks

Anomaly detection

Priority notification

COLD PATH

HAPI FHIR R4

Patient clinical records

Original observations

Longitudinal history

Clinical reports

↓

LAYER 3 — PATIENT CONTEXT INTELLIGENCE

Patient Knowledge Base

Patient ID as retrieval key

Medical history

Conditions & procedures

Prescriptions & allergies

Discharge & recovery plan

Shared Clinical Knowledge Base

Clinical context and rule-based retrieval

Clinical knowledge

Physiological relationships

Encoded tacit knowledge

Recovery & safety protocols

↓

Expert 1 — Patient Context Expert

Identifies the patient using Patient ID, retrieves the patient record, understands what the patient has undergone, and determines applicable clinical context.

Context Output

Condition · Procedure · Recovery phase · Relevant knowledge · Personalization · Unknown information

↓

LAYER 4 — PHYSIOLOGICAL EXPERT INTELLIGENCE

Expert 2 — HR

Cardiovascular intelligence

Skills + applicable tacit knowledge

Expert 3 — RR

Respiratory intelligence

Skills + applicable tacit knowledge

Expert 4 — SpO_2

Oxygenation intelligence

Skills + applicable tacit knowledge

Expert 5 — Temperature

Thermoregulatory intelligence

Skills + applicable tacit knowledge

Shared Physiological Knowledge

Context-conditioned relationships, rules, mental models and physiological constraints

Clinical Scoring & Personalized Baseline

Partial NEWS2 · Baseline deviation · Persistence · Cross-vital evidence

LAYER 5 — VIRTUAL PHYSIOLOGICAL REPRESENTATION

Textual Physiological State Manifest

Machine-readable virtual state — no images or anatomical models

Observed physiological state	Personalized baseline state
Physiological relationships	Expert evidence references
Applicable ODE configuration	Estimated model parameters
Missingness & uncertainty	State checkpoint & version

Only opaque patient/session identity and a context reference are retained. Full patient history remains in the Patient KB.

LAYER 6 — PHYSICS-INFORMED DIGITAL TWIN

Manifest

Virtual physiological state

Patient Context

Relevant numerical context features

Context-Adaptive Vector Encoder

264-dimensional feature representation with availability masks

Initial State Estimation

Reconstruct estimated current physiology

Parameter Estimation

Context-conditioned physiological parameters

↓

Hybrid PINN-ODE Core

Mechanistic Physiological ODEs

Applicable physiological relationships

+

Learned Neural Residual Dynamics

Learn dynamics not captured by simplified equations

↓

Differentiable ODE Solver

Continuous physiological state evolution

↓

State Reconciliation & Forecast Decoder

Observed state · Estimated state · Predicted trajectories · Uncertainty

5 min	**15 min**	**30 min**
Primary	*Secondary*	*Experimental*

Feedback loop: new observations → manifest update → state reconciliation → revised forecast.

↓

LAYER 7 — TRUSTWORTHY INTELLIGENCE

Uncertainty Engine

Measurement, parameter and prediction uncertainty

Counterfactual Engine

Manifest branching and evidence robustness

Trust Engine

Source, knowledge and model reliability

Evidence Engine

Grounding, provenance and deduplication

Physiological Consistency

Equation applicability and plausible bounds

Model Validity Gate

OOD detection, eligibility and abstention

↓

LAYER 8 — HYBRID DECISION INTELLIGENCE

Evidence-Constrained Decision System

Patient context + Clinical scoring + Expert evidence + Baseline deviation + Verified forecasts + Uncertainty + Counterfactual robustness

Independent Clinical Safety Supervisor

Direct hot-path input · Mandatory escalation floor · No neural dependency

Watch | **Review** | **Urgent**

Structured clinical recommendation, supporting evidence and required human action

↓

LAYER 9 — ACTION & RECOVERY INTELLIGENCE

Expert 6 — Medication & Intervention

Prescribed care, clinical review options, preparation and clinician-controlled intervention

Expert 7 — Patient Recovery

Daily activities, approved exercises, symptoms, recovery milestones and progress

Both experts receive relevant patient context directly. Neither autonomously prescribes or authorizes treatment.

↓

LAYER 10 — HUMAN INTERACTION & RECORDS

Clinician Command Center

Live vitals, twin, expert evidence, alerts, forecasts and clinical review

Patient Recovery Assistant

Daily recovery, approved care plan, reminders and symptom check-in

Hospital Record Interface

FHIR observations, clinical reports and longitudinal history

Evaluation & Security Console

Simulation, replay, benchmarking, data quality and security audit

CROSS-CUTTING GOVERNANCE

Privacy · Cybersecurity · Clinical safety · Patient isolation · Human oversight · Evidence provenance · Performance · Auditability

---

2. Patient Knowledge Base and Shared Clinical Knowledge Base

This is the final knowledge architecture.

We have two distinct knowledge bases because patient-specific facts and general clinical knowledge serve different purposes.

2.1 Patient Knowledge Base — Patient KB

The Patient KB stores or references all relevant information about an individual patient.

Primary lookup key: Patient ID

Information category	Contents
Patient identity	Internal patient ID, FHIR identity mapping, demographics
Clinical history	Documented medical conditions and encounters
Procedures	Verified surgeries and treatments undergone
Medication history	Prescriptions, reported current usage and adherence
Discharge information	Discharge encounter, condition and date
Recovery context	Approved recovery plan, restrictions and milestones
Historical physiology	Previous observations and clinical measurements
Personalization	Authorized oxygen scale, individual targets and relevant constraints
Safety information	Documented allergies, contraindications and restrictions
Care team	Assigned clinician and authorized backup

The existing Clinical Registry V1 and HAPI FHIR records provide the authoritative patient information.

The Patient KB can maintain an indexed, runtime-safe representation of those records instead of duplicating the entire clinical database.

Example Patient KB record

```
{
  "patient_id": "P001",
  "record_version": 3,
  "clinical_history": {
    "documented_conditions": [
      {
        "condition": "Pneumonia",
        "evidence_ref": "Condition/COND-001"
      }
    ],
    "verified_procedures": [],
    "allergy_status": "unknown"
  },
  "discharge_context": {
    "encounter_ref": "Encounter/ENC-001",
```

```
    "primary_condition": "Pneumonia",
    "recovery_phase": "early"
  },
  "medication_context": {
    "prescription_refs": [
      "MedicationRequest/MED-001"
    ],
    "current_adherence": "unknown"
  },
  "recovery_plan_ref": "CarePlan/CP-001",
  "personalization": {
    "oxygen_scale": 1,
    "baseline_profile_ref": "BASELINE-P001"
  },
  "care_team_ref": "CareTeam/TEAM-001"
}
```

This is an illustrative synthetic record, not an extracted patient record.

2.2 Shared Clinical Knowledge Base — Clinical KB

This contains general knowledge that can be reused across different patients.

Knowledge category	Examples
Explicit clinical knowledge	NEWS2 components, measurement interpretation and clinical guidelines
Physiological knowledge	Cardiorespiratory coupling, thermoregulation and recovery dynamics
Encoded tacit knowledge	Contextual reasoning heuristics and physiological mental models
Medication knowledge	Known medication effects, interactions and limitations
Recovery knowledge	Condition-specific recovery protocols and activity restrictions
Physiological modeling	ODE equations, parameter bounds and applicability
Safety knowledge	Criticality rules, verification and escalation policies

Each knowledge entry should have a rule ID, version, applicability criteria, supporting references, exceptions and limitations.

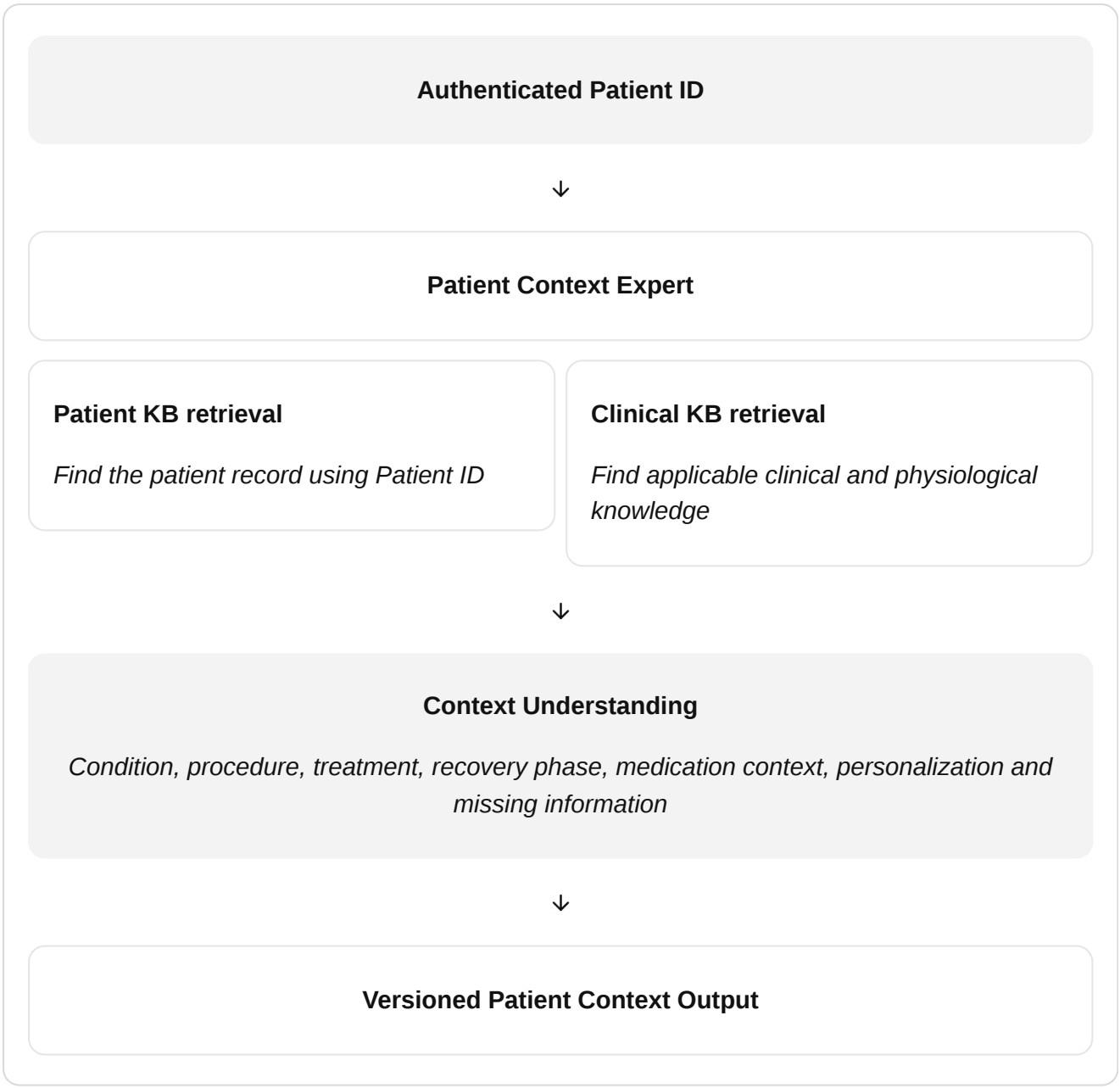
The shared KB is not patient-specific. The Patient Context Expert determines which knowledge is applicable to a particular patient.

---

3. Expert 1 — Patient Context Expert

This is the **context controller of the entire intelligence system**.

Its responsibility is to identify the patient and understand what the patient has undergone so that every downstream component can work appropriately.



Patient Context Expert workflow

1. Receive the authenticated Patient ID.
2. Retrieve the corresponding record from the Patient KB.
3. Identify documented conditions, verified procedures and treatments.
4. Determine the patient's recovery phase and approved care requirements.
5. Retrieve applicable knowledge from the Shared Clinical KB.
6. Produce a structured context for the downstream intelligence components.

Example context output

```
{
  "context_id": "CTX-P001-V3",
  "patient_id": "P001",
  "patient_record_version": 3,

  "clinical_context": {
    "documented_condition": "Pneumonia",
    "clinical_domain": "respiratory",
    "recovery_phase": "early"
  },

  "applicable_knowledge_refs": [
    "RESPIRATORY_RECOVERY_V1",
    "CARDIORESP_COUPLING_V1",
    "OXYGENATION_INTERPRETATION_V1"
  ],

  "personalization": {
    "oxygen_scale": 1,
    "baseline_profile_ref": "BASELINE-P001"
  },

  "model_context": {
    "equation_set": "CARDIORESP_THERMAL_V1",
    "parameter_profile_ref": "PARAM-P001-V1"
  },

  "unknown_information": [
    "current_medication_adherence"
  ]
}
```

This context is provided to the four physiological experts, encoder, decision system, medication expert and recovery expert.

The manifest does not need to repeat this clinical history.

---

4. Experts 2–5 — Physiological Expert Intelligence

These four experts operate in parallel after receiving the Patient Context Expert’s output.

They remain general physiological experts rather than separate disease-specific agents.

Heart Rate Expert

Cardiovascular intelligence

HR trends and baseline deviations

Cardiovascular compensation

Medication-related influences

Activity and recovery response

Cross-vital relationships

Respiratory Rate Expert

Respiratory intelligence

Respiratory trends

Sustained respiratory changes

Compensatory responses

Activity-related variation

Respiratory recovery behavior

SpO_2 Expert

Oxygenation intelligence

Oxygen saturation trends

Personalized oxygen interpretation

Desaturation persistence

Probe artifact recognition

Cardiorespiratory evidence

Temperature Expert

Thermoregulatory intelligence

Temperature trends

Thermal baseline deviations

Fever-related physiological changes

Recovery context

Thermal–cardiovascular relationships

How context changes expert behavior

Patient context	Expert adaptation
Pneumonia recovery	Respiratory and oxygenation relationships become particularly relevant
Verified cardiac procedure	Cardiovascular recovery and applicable restrictions receive attention
Historical beta-blocker prescription	Consider possible blunted HR response, without assuming current use
Authorized hypercapnic respiratory failure	Apply appropriate oxygen interpretation
Verified orthopedic procedure	Consider approved mobility restrictions and activity response

All four experts remain active. Context determines applicable knowledge and interpretation, not which vital signs can be ignored.

Expert outputs

Each expert produces structured findings containing:

Output	Meaning
Clinical component score	Applicable partial NEWS2 component
Contextual concern	None, mild, moderate or high
Baseline deviation	Difference from previous eligible baseline
Trend slope	Rate of physiological change
Abnormal duration	Observed persistence of an abnormality
Cross-signal support	Supporting evidence from other vital domains
Evidence reliability	Reliability of the underlying evidence
Interpretation uncertainty	Uncertainty in the expert's interpretation
Evidence references	Original measurements and clinical context
Knowledge references	Applied rules and versions

The experts also identify relevant physiological relationships that can inform the virtual manifest and PINN-ODE.

These relationships are hypotheses or applicable modeling assumptions unless sufficiently supported by evidence.

---

5. Clinical scoring and personalized baseline intelligence

The scoring and baseline modules support the physiological experts and the final decision system.

They are deterministic clinical intelligence modules, not additional agents.

Clinical scoring

The system computes component-level scores for HR, RR, SpO_2 and temperature using NEWS2-style rules.

Important correction to the earlier architecture: Standard NEWS2 temperature scoring includes 3 points for temperature at or below 35°C. Therefore, the maximum possible four-component subtotal is **12**, not 11.

Component	Maximum score
Heart rate	3
Respiratory rate	3
SpO_2	3
Temperature	3
Four-component subtotal	**12**

The full NEWS2 system also includes systolic blood pressure, consciousness and supplemental oxygen scoring.

Our four-component subtotal must therefore be clearly identified as partial NEWS2, and full NEWS2 escalation thresholds must not be directly applied to it.

Personalized baseline

The baseline engine maintains an independent physiological reference for each vital.

1. **Historical reference** Retrieve available historical observations from the Patient KB.
2. **Calibration** Establish an individual baseline from eligible, stable and reliable observations.
3. **Established baseline** Compare incoming observations against the previously established baseline.
4. **Freeze** Stop baseline adaptation during suspected deterioration or unreliable measurements.
5. **Resume** Resume adaptation only after the applicable stability and eligibility requirements are satisfied.

The baseline must not gradually adapt to a deteriorating physiological state and incorrectly classify that deterioration as the patient's new normal.

---

6. Textual Physiological State Manifest

This is the first part of our combined digital twin.

What the manifest actually is

The manifest is a **structured textual representation of the virtual physiological state of a patient**.

It contains the information required to reconstruct, model, update and audit the patient's computational physiological state.

It is not a patient biography, clinical database, image or anatomical reconstruction.

What belongs inside the manifest

Component	Purpose
Opaque patient/session identity	Associate the state with the correct patient and session
Context reference	Reference the Patient Context Expert's output
Observed physiological state	Current measured HR, RR, SpO_2 and temperature
Baseline state	Individual physiological reference values and freeze status
Physiological relationships	Applicable relationships and expert-supported findings
Model configuration	Applicable ODE equations and parameter definitions
Estimated parameters	Current patient-conditioned model parameters
Quality and missingness	Measurement reliability and unavailable information
Expert evidence references	Supporting physiological expert findings
State checkpoint	Reconstructable computational state
Version and time	Reproduce the state at a particular physiological time

What does not belong inside the manifest

Full patient medical history, patient name, detailed diagnosis descriptions, complete surgical records, medication prescription history, personal demographic records, images, 3D anatomy or hidden simulator ground truth.

Those remain in their respective authoritative sources.

Final manifest example

```
{
  "manifest_id": "MANIFEST-P001-0042",
  "schema_version": "1.0",
  "state_version": 42,

  "identity": {
    "patient_id": "P001",
    "session_id": "SESSION-001",
    "variant_id": "original"
  },

  "source_time_s": 630,

  "context_ref": "CTX-P001-V3",

  "observed_state": {
    "hr": {
```

```
      "value": 108,
      "unit": "bpm",
      "quality": "valid",
      "observation_ref": "OBS-HR-630"
    },
    "rr": {
      "value": 26,
      "unit": "breaths/min",
      "quality": "valid",
      "observation_ref": "OBS-RR-630"
    },
    "spo2": {
      "value": 92,
      "unit": "%",
      "quality": "valid",
      "observation_ref": "OBS-SPO2-630"
    },
    "temperature": {
      "value": 38.1,
      "unit": "Cel",
      "quality": "valid",
      "observation_ref": "OBS-TEMP-630"
    }
  },

  "baseline_state": {
    "hr": 78,
    "rr": 16,
    "spo2": 97,
    "temperature": 37.0,
    "adaptation_status": "frozen",
    "baseline_ref": "BASELINE-P001-V4"
  },

  "physiological_relationships": [
    {
      "relationship": "cardiorespiratory_coupling",
      "status": "supported",
      "evidence_ref": "EVIDENCE-42-A"
    },
    {
      "relationship": "thermal_cardiovascular_coupling",
      "status": "possible",
      "evidence_ref": "EVIDENCE-42-B"
    }
  ],

  "model_configuration": {
    "equation_set": "CARDIORESP_THERMAL_V1",
    "parameter_profile_ref": "PARAM-P001-V2",
    "parameter_status": "provisional"
  },

  "expert_evidence_refs": [
    "HR-E42",
```

```
    "RR-E42",
    "SPO2-E42",
    "TEMP-E42"
  ],

  "uncertainty": {
    "missing_vitals": [],
    "parameter_uncertainty": "moderate",
    "unobserved_physiology": "unknown"
  },

  "estimated_state_ref": "ESTIMATE-P001-42",
  "checkpoint_ref": "CHECKPOINT-P001-42",
  "parent_manifest_ref": "MANIFEST-P001-0041"
}
```

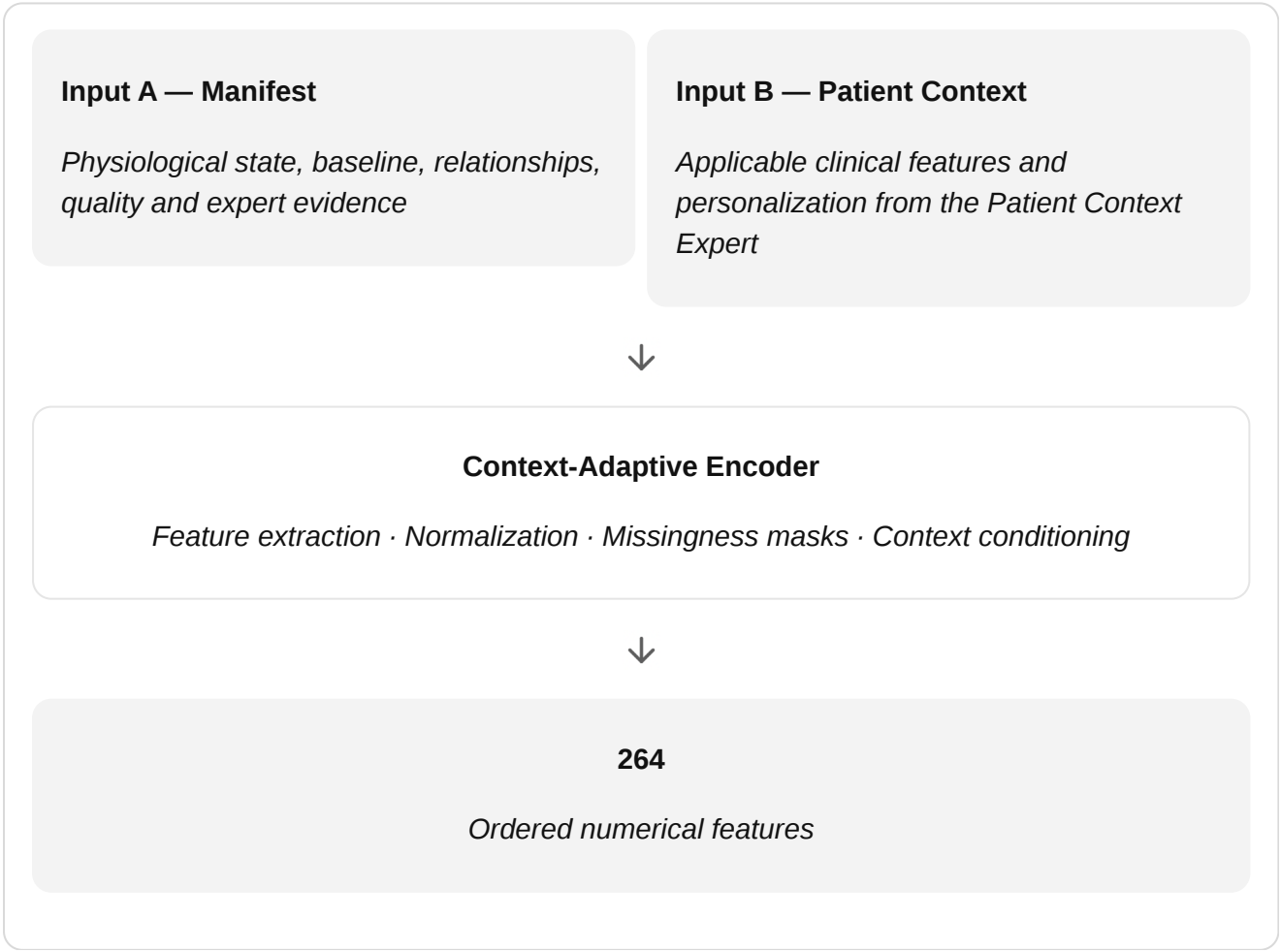

This is the proposed structure for an illustrative synthetic state.

The Patient Context Expert's output is referenced through `context_ref`. There is no duplication of the patient's entire clinical record.

---

7. Context-adaptive vector encoder

The encoder receives two distinct inputs:



↓

PINN-ODE State & Parameter Estimation

Final encoder feature structure

Feature group	Dimensions
Physiological features	48
Physiological availability masks	48
Signal quality features	24
Expert-derived features	32
Expert feature masks	32
Patient context features	32
Context availability masks	32
Temporal and baseline features	16
Total	**264**

The encoder preserves missingness explicitly.

A missing temperature value must not become a real temperature measurement of zero.

The initial model uses a fixed 15-second source-time feature timeline, even when the gateway uses adaptive 5/10/15-second windows. Immediate safety evaluation remains event-driven.

---

8. Manifest + PINN-ODE Cognitive Digital Twin

This is the central innovation of the architecture.

Manifest = representation of the current virtual physiological state.

PINN-ODE = mathematical evolution of that state.

The combined system reconstructs an estimated patient state, models its dynamics, forecasts possible changes and continuously updates itself as new observations arrive.

8.1 Internal architecture

Hybrid Physics-Informed Twin

Encoded Manifest + Context

↓

State Estimator

Estimate the current four-vital physiological state

Parameter Estimator

Estimate constrained patient-specific parameters

↓

Coupled Physiological Dynamics

HR dynamics

Cardiovascular state evolution

RR dynamics

Respiratory state evolution

SpO_2 dynamics

Oxygenation state evolution

Temperature dynamics

Thermoregulatory state evolution

+

Shared Context-Conditioned Neural Residual

Learn additional physiological dynamics from training trajectories

↓

Differentiable ODE Solver

Integrates the coupled state over continuous physiological time

↓

Forecast Decoder & Uncertainty Estimation

↓

Predicted Physiological Trajectories

HR · RR · SpO₂ · Temperature

8.2 Mathematical architecture

Let the Patient Context Expert produce:

$$C_t = ContextExpert(PatientID, KB)$$

The four experts produce:

$$E_t = Experts(X_t, C_t, K)$$

The textual manifest is constructed as:

$$M_t = Manifest(X_t, E_t, B_t, Q_t)$$

Where X_t represents observations, B_t represents baseline state and Q_t represents quality and uncertainty.

The encoder combines the manifest and clinical context:

$$z_t = Encoder(M_t, C_t)$$

The physiological state is:

$$x(t) = \begin{bmatrix} HR(t) \\ RR(t) \\ SpO_2(t) \\ T(t) \end{bmatrix}$$

The final hybrid PINN-ODE formulation is:

$$\boxed{\frac{dx}{dt} = f_{\text{phys}}(x, p, u) + f_\theta(x, z_t, u)}$$

Term	Meaning
f_{phys}	Simplified mechanistic physiological dynamics

Term	Meaning
f_θ	Learned neural residual dynamics
p	Constrained patient-specific physiological parameters
u	Available external influences, such as reported activity
z_t	Context-conditioned encoded manifest
$x(t)$	Current estimated physiological state

8.3 Example physiological equation

An illustrative HR equation is:

$$\frac{dHR}{dt} = -\frac{HR - HR_b}{\tau_{HR}} + \alpha_T(T - T_b) + \alpha_A A(t) + r_\theta$$

The equation represents baseline recovery dynamics, temperature influence, activity influence and learned residual behavior.

The exact physiological equations, coefficients, units and parameter bounds must be agreed upon by the MATLAB, clinical and AI teams.

This equation is an engineering example, not a clinically validated individual heart-rate prediction law.

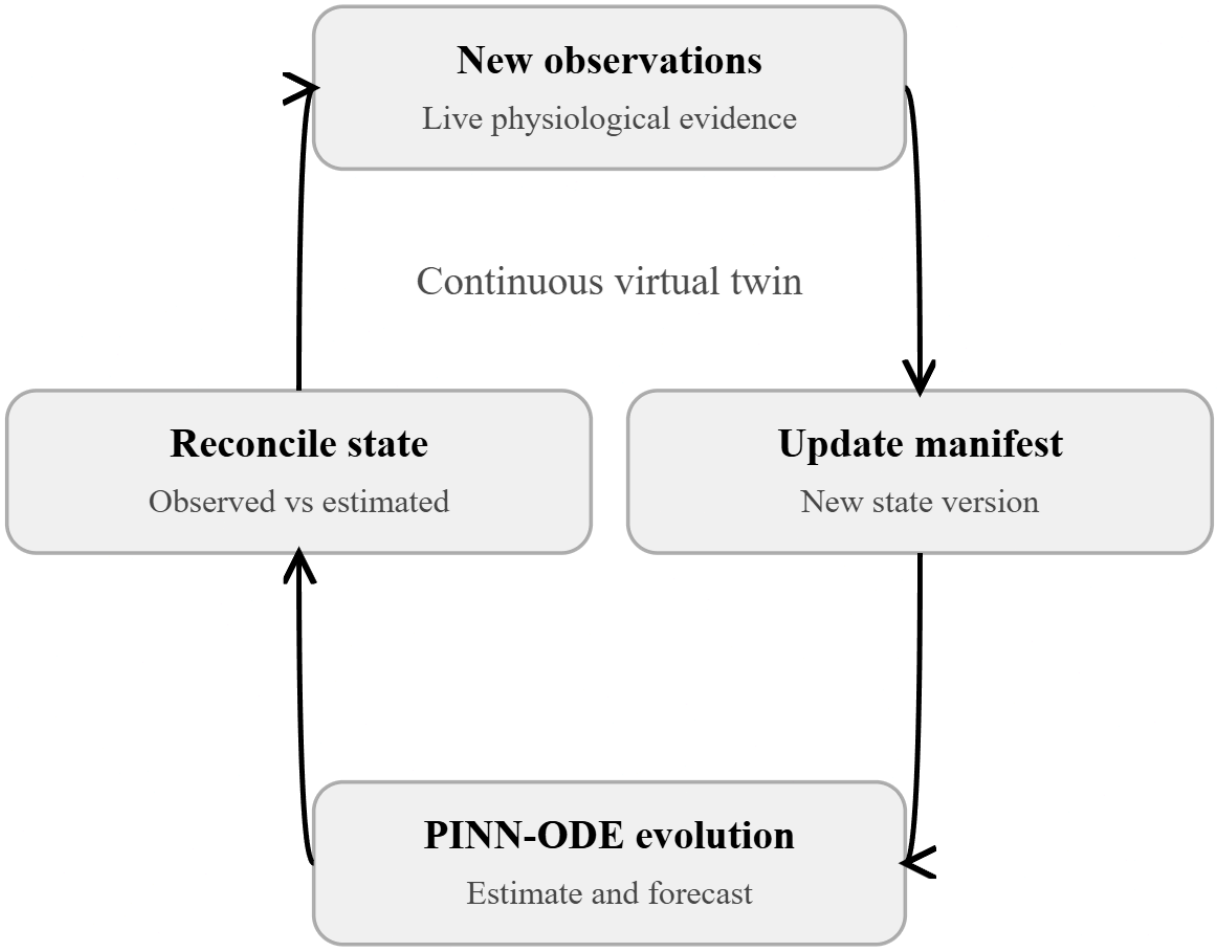
8.4 Physics-informed training

$$\mathcal{L}_{\text{total}} = \lambda_d \mathcal{L}_{\text{data}} + \lambda_p \mathcal{L}_{\text{physics}} + \lambda_i \mathcal{L}_{\text{initial}} + \lambda_b \mathcal{L}_{\text{bounds}} + \lambda_r \mathcal{L}_{\text{residual}}$$

The training objectives ensure that the model learns from available trajectories while respecting applicable physiological constraints and plausible parameter ranges.

Clinical targets are not automatically physical constraints. For example, a patient's authorized SpO_2 target must not force the predicted oxygenation state to remain within that target.

8.5 Continuous adaptation



Patient-specific state and parameters adapt to incoming evidence. Shared trained neural weights remain fixed during normal runtime.

The model does not require retraining for every new patient observation.

Personalization occurs through context, initial-state estimation, baseline state, constrained parameters and observation reconciliation.

---

9. Trustworthy Intelligence Layer

This layer verifies the intelligence generated by the physiological experts and digital twin before it contributes to the final decision.

9.1 Uncertainty Engine

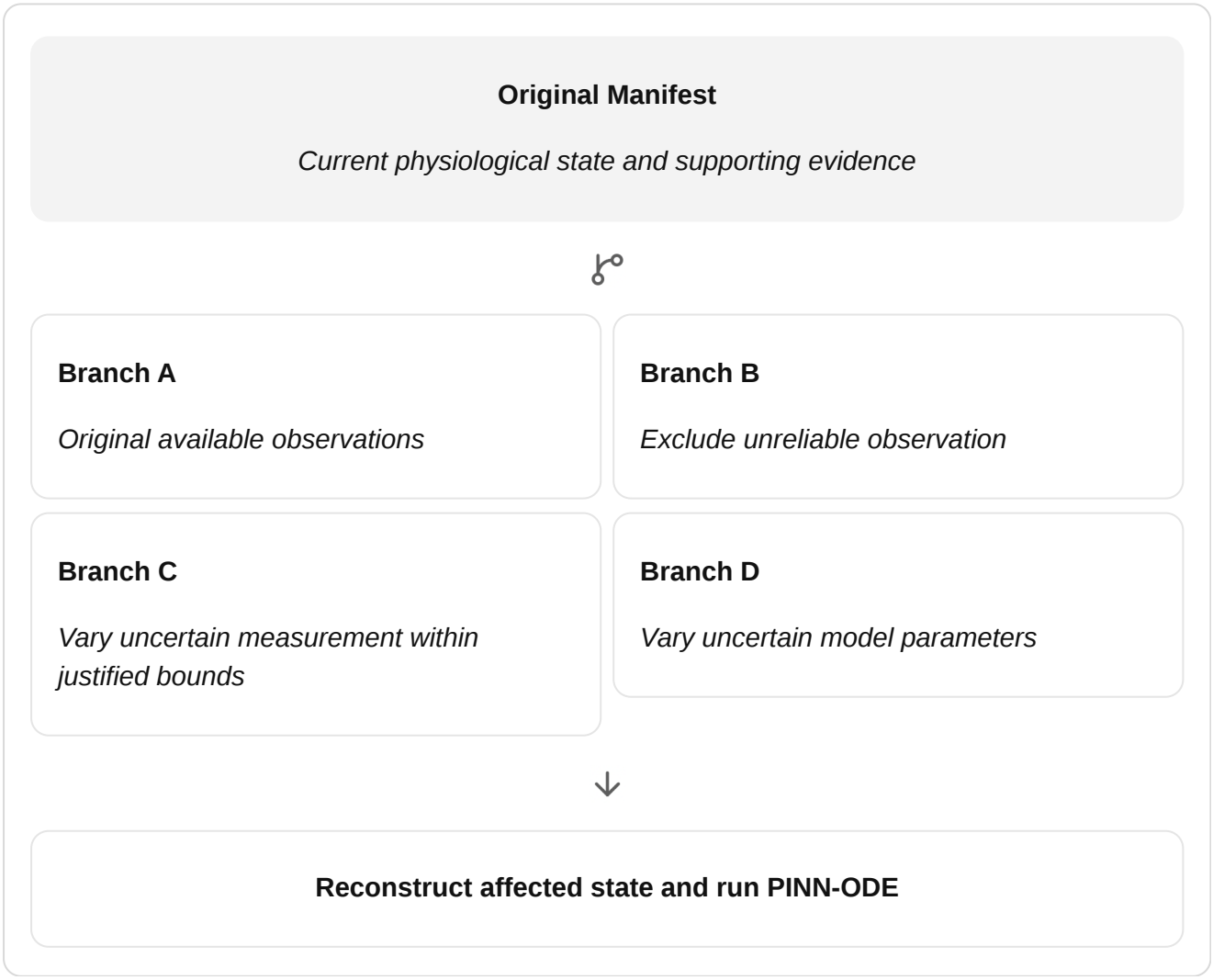
Identifies and represents different types of uncertainty.

Type	Example
Measurement uncertainty	Unreliable oxygen sensor
Context uncertainty	Unknown medication adherence
Baseline uncertainty	Insufficient calibration
Parameter uncertainty	Poorly identifiable physiological parameter
Model uncertainty	Unseen physiological behavior
Forecast uncertainty	Wide future prediction interval
Interpretation uncertainty	Multiple plausible explanations

The system must be able to abstain from a forecast when evidence is insufficient.

9.2 Counterfactual Engine

The manifest enables isolated virtual experiments.



↓

Compare forecast divergence and decision robustness

The original patient state remains unchanged.

These experiments test evidence and model sensitivity. They do not establish the causal effect of administering a medication or performing a treatment.

9.3 Trust Engine

Evaluates source authenticity, signal quality, knowledge applicability, model validity, patient-state integrity and forecast reliability.

9.4 Evidence Engine

Every recommendation must be traceable to actual observations, patient context, applicable knowledge, expert findings and relevant model outputs.

Overlapping expert findings must be deduplicated so that multiple interpretations of the same physiological evidence are not incorrectly treated as independent confirmations.

9.5 Physiological Consistency and Model Validity

Checks numerical stability, applicable physiological relationships, plausible output bounds, out-of-distribution conditions and forecast eligibility.

Experimental or unvalidated forecasts can be displayed for evaluation but cannot independently change clinical priority.

---

10. Hybrid Decision Intelligence

The decision system combines all verified evidence.

It is not dependent on a single neural risk score.

$$D_t = F(C_t, S_t, E_t, B_t, P_t, U_t, R_t)$$

Where:

Input	Meaning
C_t	Patient clinical context
S_t	Clinical scoring

Input	Meaning
E_t	Physiological expert evidence
B_t	Personalized baseline deviation
P_t	Eligible PINN-ODE forecasts
U_t	Uncertainty
R_t	Trust and counterfactual robustness

Decision hierarchy

1. **Independent safety evaluation** Determine whether the available evidence establishes a mandatory escalation requirement.
2. **Current physiological assessment** Analyze actual measurements, clinical scoring, baseline deviations and expert findings.
3. **Predictive assessment** Consider eligible PINN-ODE forecasts and associated uncertainty.
4. **Evidence verification** Resolve conflicts, evaluate reliability and preserve unknown information.
5. **Clinical recommendation** Determine Watch, Review or Urgent using the applicable decision policy.
6. **Human action** Generate an evidence report, notify the appropriate clinician and support approved care.

Final decision output

Dimension	Output
Clinical priority	Watch / Review / Urgent / Insufficient assessment
Physiological concern	None / Mild / Moderate / High / Unknown
Evidence reliability	Low / Moderate / High / Unknown
Monitoring status	Normal / Degraded / Interrupted
Forecast status	Validated / Experimental / Abstained / Failed
Recommended action	Continue monitoring / Verify / Review / Escalate
Supporting evidence	Observations, clinical rules, context and expert findings
Limitations	Missing information, uncertainty and model limitations

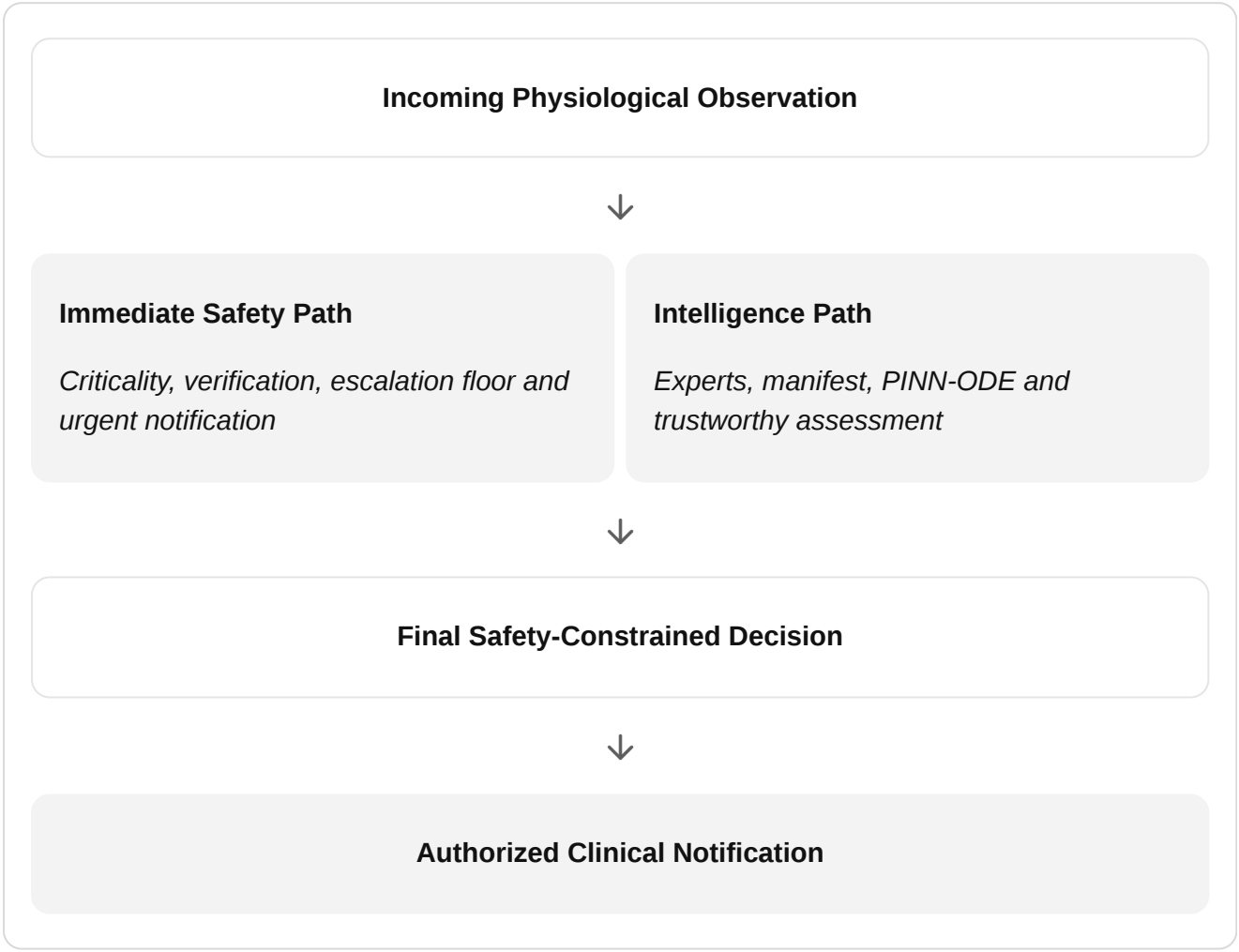
Physiological concern and evidence reliability remain separate.

An unreliable measurement indicating potentially serious deterioration must not be automatically dismissed.

---

11. Independent clinical safety path

This is a mandatory parallel architecture, not simply the last step of the neural pipeline.



The immediate safety path remains functional during missing data, baseline calibration, PINN-ODE failure, unreliable forecasts, gateway overload and HAPI persistence delays.

Urgent notifications must not wait for the neural model or detailed report.

The final decision cannot downgrade a mandatory safety requirement merely because the PINN-ODE predicts recovery.

---

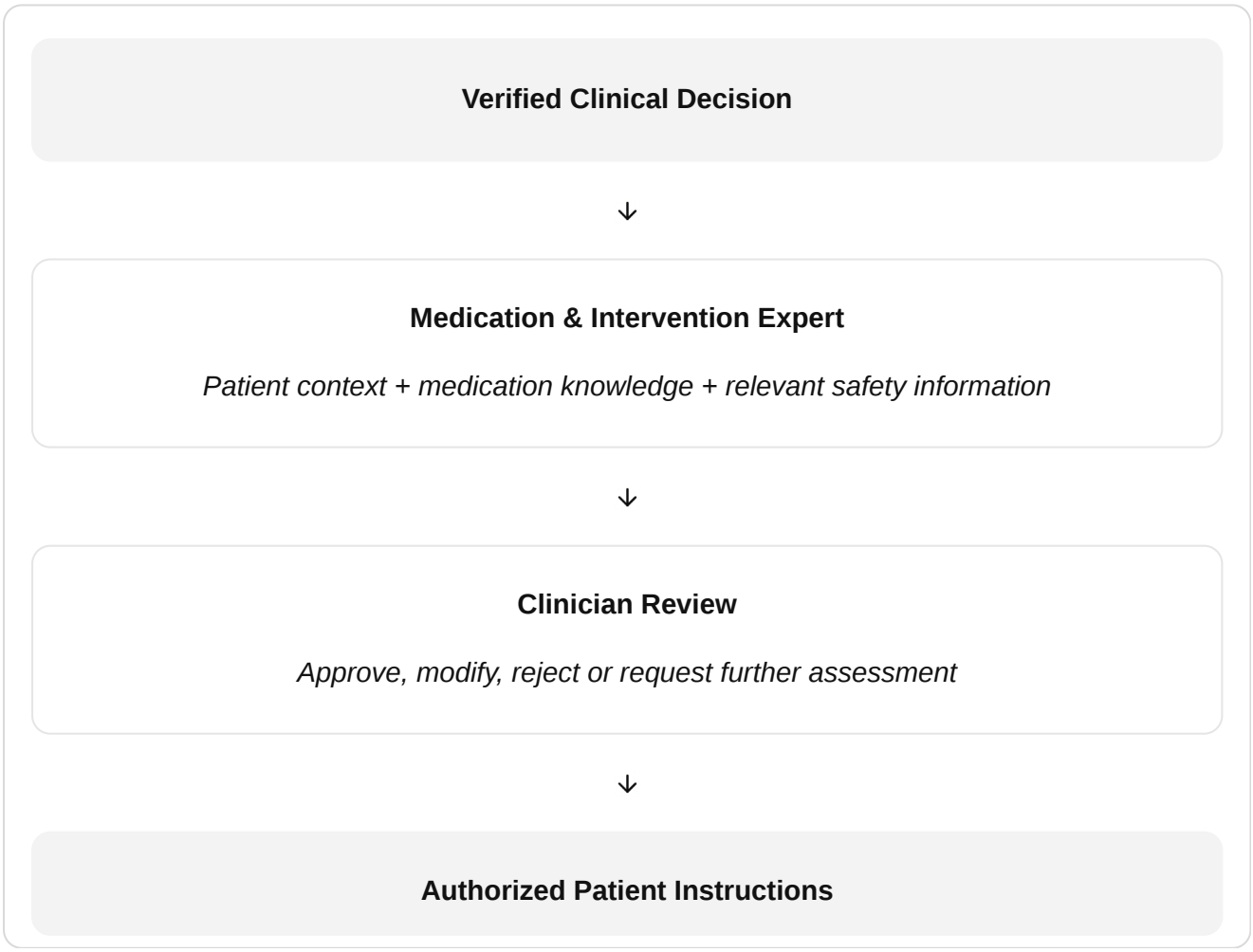
12. Expert 6 — Medication & Clinical Intervention Expert

This expert receives the Patient Context Expert's output, medication knowledge, verified decision and relevant clinical evidence.

It can prepare contextual information in parallel, but intervention recommendations must wait for the verified clinical assessment.

Patient situation	Expert response
Stable recovery	Continue existing approved care
Prescribed medication due	Generate an appropriate reminder
Possible medication-related concern	Present supporting evidence for clinician review
Potential intervention	Provide evidence-backed options to the clinician
Urgent deterioration	Support immediate escalation and clinical preparation
New treatment instruction	Require authorized clinician approval

Human-supervised workflow



The expert does not autonomously prescribe new medication, change dosage or authorize procedures.

---

13. Expert 7 — Patient Recovery Expert

This expert continuously supports the patient's recovery journey.

It retrieves relevant patient context and approved care-plan information through the Patient Context Expert.

Recovery intelligence inputs

Input	Purpose
Approved recovery plan	Establish applicable recovery goals
Patient activities	Understand activity tolerance and progress
Symptoms	Identify new or worsening concerns
Reported medication usage	Track reported adherence
Physiological trends	Understand response during recovery
Approved exercises	Monitor task completion and tolerance
Verified clinical decision	Adjust monitoring and escalation behavior

Recovery timeline

1. **7 days — Immediate recovery** Approved medication reminders, rest, symptom monitoring and early recovery tasks.
2. **15 days — Early recovery** Activity tolerance, progress assessment and follow-up preparation.
3. **30 days — Functional recovery** Approved activity progression and condition-specific milestones.
4. **90 days — Rehabilitation** Longitudinal functional progress and approved rehabilitation goals.
5. **180 days — Long-term recovery** Review of longer-term recovery goals and follow-up requirements.

These are possible care-plan durations, not mandatory monitoring periods for every patient.

The Recovery Expert provides daily summaries, approved tasks, symptom check-ins, medication reminders and progress monitoring.

It does not independently authorize exercise progression during unresolved clinical concerns.

---

14. Final dashboard architecture

The dashboard must expose the intelligence generated by the architecture in a meaningful way.

The clinician interface provides technical and clinical evidence. The patient interface provides understandable recovery support.

A. Clinician Command Center

Clinical Command Center **Demo data**

Illustrative interface

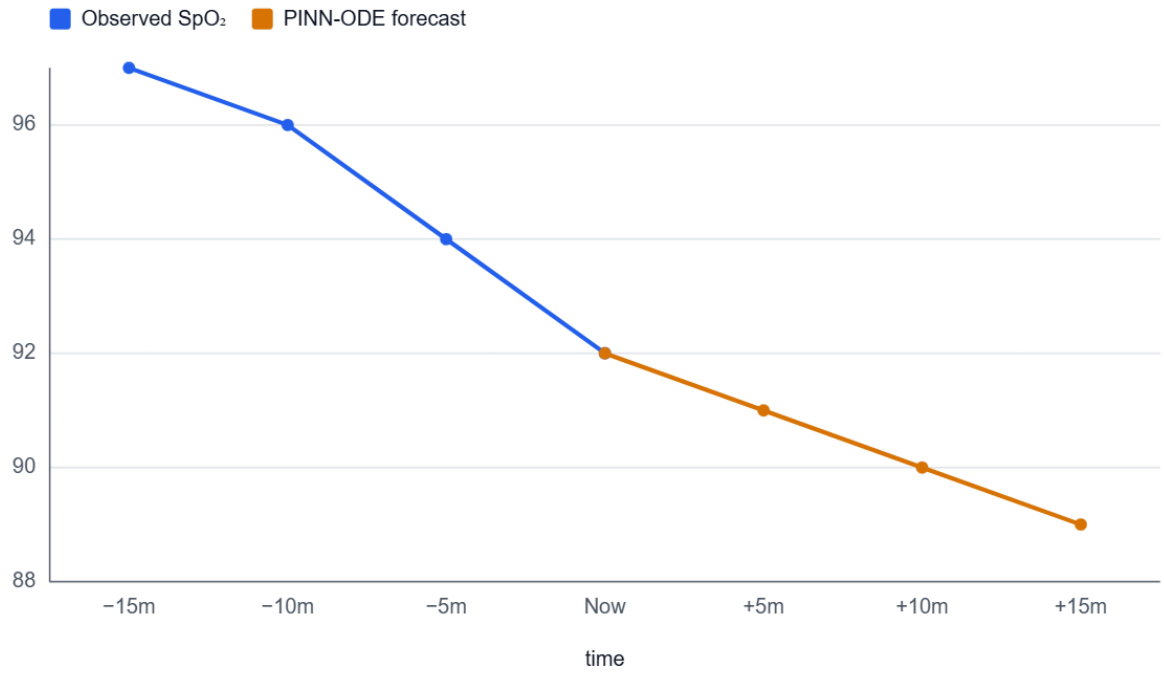
Monitored	*Review*	*Urgent*
10	**2**	**1**

Patient P001 — Live Digital Twin

Post-discharge monitoring · Synthetic example

Heart rate **108** · bpm *Baseline: 78*	*Respiratory rate* **26** · /min *Baseline: 16*
SpO_2 **92** · % *Baseline: 97*	*Temperature* **38.1** · °C *Baseline: 37.0*

Physiological trajectory



time	Observed SpO₂	PINN-ODE forecast
−15m	97	
−10m	96	
−5m	94	
Now	92	92
+5m		91
+10m		90
+15m		89

Illustrative forecast, not a result from a trained or validated model.

Expert intelligence

HR Expert	Elevated relative to baseline
RR Expert	Sustained respiratory increase
SpO₂ Expert	Declining oxygenation
Temperature Expert	Increasing temperature

Trustworthy assessment

Physiological concern
High

Evidence reliability
High

Baseline adaptation
Frozen

Forecast status
Experimental

Clinical recommendation

Review the sustained physiological changes, original observations and applicable patient context. Obtain an authorized clinical assessment.

Final clinician dashboard pages

Page	Main functionality
Overview	Patient queue, clinical priority and monitoring status
Patient Profile	Clinical information retrieved from Patient KB
Live Monitoring	Four vital signs, trends, baselines and signal quality
Cognitive Digital Twin	Textual manifest, estimated state, parameters and forecasts
Expert Intelligence	Four expert findings and supporting clinical knowledge
Trustworthy Intelligence	Uncertainty, evidence, counterfactuals and model validity
Alert Center	Clinical and technical alerts, acknowledgment and escalation
Clinical Reports	Evidence-grounded assessments and recommendations
Medication Review	Existing prescriptions and clinician-controlled intervention
Recovery Progress	Daily activities, symptoms and recovery milestones
Patient History	FHIR records and longitudinal clinical timeline

The digital twin page can visualize numerical trends and forecasts. That does not mean the manifest itself contains images or anatomical representations.

B. Patient Recovery Assistant

My Recovery



Day 4 · Illustrative care plan

Today's recovery tasks

2 of 4

Daily progress

Today's approved care plan

☐ Morning prescribed medication
☐ Approved walking activity
☐ Daily symptom check-in
☐ Evening prescribed medication

Recovery phase

Early recovery

Reported walking

8 min

Recovery guidance

Follow your approved care plan and report new or worsening symptoms to your care team.

Interactive mockup only. Task changes are not transmitted to a clinician.

C. Evaluation and Security Console

This interface contains:

Module	Functionality
Simulator controls	Select patient, physiological scenario and replay speed
BIDMC replay	Select independent real-data recording
Data quality	Missingness, artifacts and timestamp gaps
Model evaluation	Forecast errors and uncertainty coverage
Clinical evaluation	Detection lead time and false alerts
Manifest validation	State reconstruction and reproducibility
Security monitoring	Authentication failures, replay and unauthorized access
System performance	Latency, throughput and resource usage

Module	Functionality
FHIR monitoring	Persistence, validation and reconciliation

15. Privacy, cybersecurity and patient isolation

The Patient ID is the primary Patient KB lookup key, but it is not sufficient by itself for authorization or runtime state isolation.

Identity architecture

Identity	Purpose
Patient ID	Retrieve the patient from Patient KB
FHIR Patient ID	Access authorized standardized clinical records
Source namespace	Separate Synthea, BIDMC and other sources
Variant ID	Separate original and synthetic test variants
Session ID	Separate independent monitoring or replay sessions
Manifest ID	Identify a particular physiological state version

The runtime state key should include:

```
Source Namespace
    + Patient/Subject ID
    + Variant ID
    + Session ID
```

This prevents independent recordings or simulation runs from sharing model state and personalized baselines.

Security mechanisms

Security area	Implementation
Source authentication	Authenticated simulator and data producers
Transport	TLS
Event integrity	Authenticated events and replay protection
Access control	Role-based and patient-level authorization
Data privacy	Pseudonymization and minimum necessary information
Storage	Encryption and controlled key management
State isolation	Independent patient/session state
Manifest integrity	Versioned references and tamper-evident audit
Clinical data	Restricted HAPI FHIR access

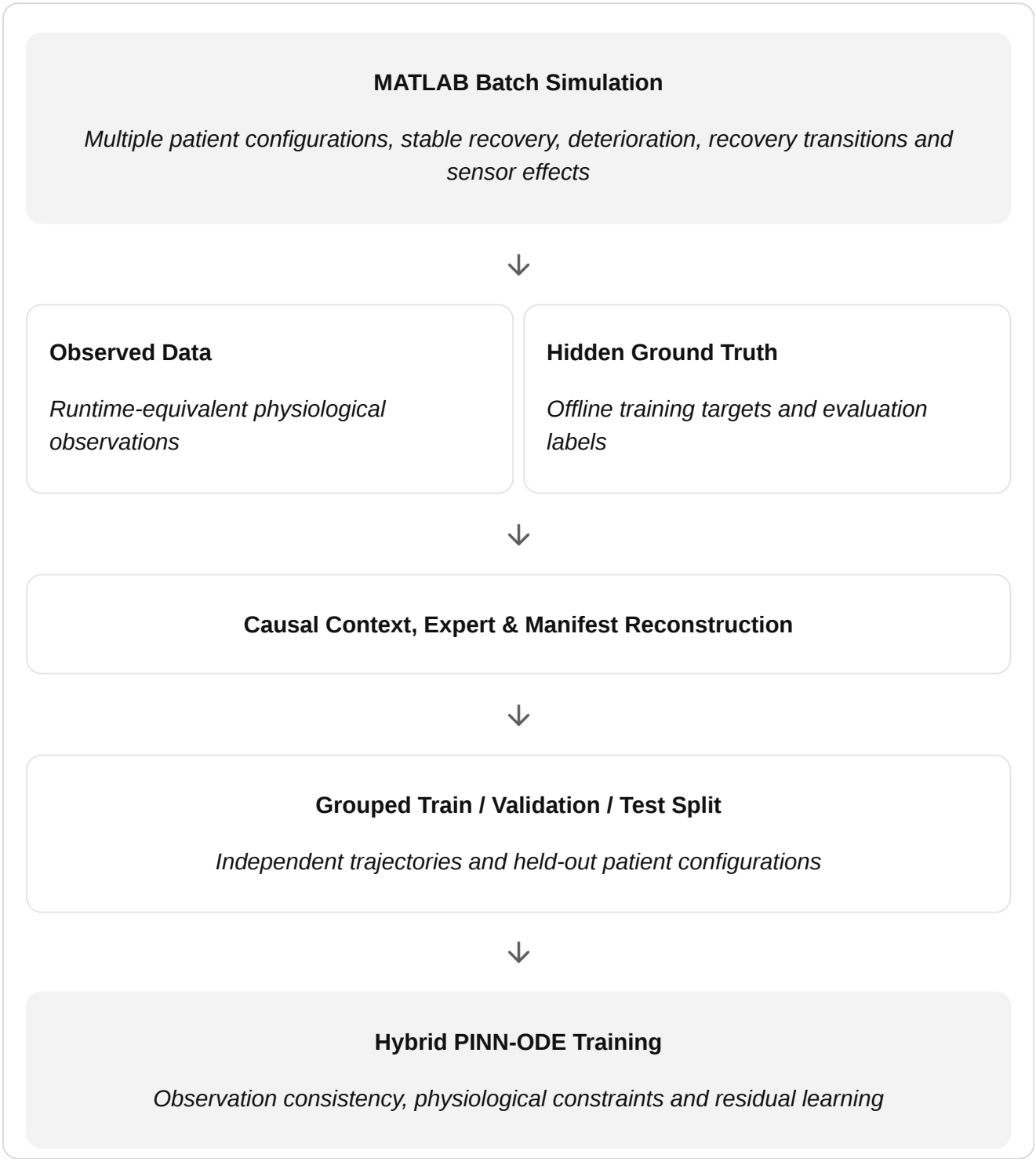
Security area	Implementation
Security audit	Access logs and security event records

The manifest's lack of duplicated personal and clinical information reduces unnecessary exposure, but physiological measurements and opaque identifiers still require appropriate privacy protection.

---

16. Training and evaluation architecture

The training pipeline remains separate from the operational clinical system.





Forecast, Clinical & System Evaluation

Evaluation requirements

Category	Metrics
Deterioration detection	Lead time, detected events, missed events
Alert performance	False alerts, alerts per patient-day
Forecasting	MAE and RMSE per vital and horizon
Uncertainty	Prediction interval coverage and width
Personalization	Improvement over non-personalized monitoring
Artifact handling	Precision, recall and missed genuine abnormalities
Missingness	Detection latency and safe abstention
Manifest	Reconstruction accuracy and reproducibility
PINN-ODE	Solver stability, physiological consistency and latency
Explainability	Evidence completeness and rule traceability
Security	Replay rejection, unauthorized access and patient isolation
System performance	Throughput, p50/p95/p99 latency and resource usage
FHIR	Resource validation and persistence

Baseline comparisons

For clinical monitoring, compare the proposed system against a naive fixed-threshold alerting system.

For physiological forecasting, compare persistence, linear trend, mechanistic ODE, unconstrained neural ODE and the hybrid PINN-ODE.

BIDMC is used for independent real-data replay and limited forecasting evaluation. It does not contain labelled post-discharge deterioration outcomes or temperature measurements.

Synthetic training performance must not be presented as clinical validation.

---

17. Final technology stack

Component	Technology
Virtual wearable	MATLAB
Real physiological replay	BIDMC PhysioNet
Patient clinical profiles	Synthea
Patient KB	Clinical Registry + HAPI FHIR + indexed context store
Shared Clinical KB	Versioned JSON/YAML and deterministic knowledge functions
Operational backend	Python + FastAPI
Healthcare interoperability	HAPI FHIR R4
Shared interfaces	Pydantic + JSON Schema
Textual manifest	Structured JSON
Feature processing	NumPy + SciPy
Neural modeling	PyTorch
Differentiable ODE solver	torchdiffeq
Physics-informed training	PyTorch automatic differentiation
Operational state store	PostgreSQL or SQLite for prototype
Durable event processing	Redis Streams or equivalent
Live dashboard communication	WebSockets
Frontend	React + TypeScript
Data visualization	Recharts or Plotly
Deployment	Docker Compose
Testing	Pytest
Performance evaluation	Replay benchmarks and load testing

18. Final shared contracts

These contracts define how the components communicate without duplicating responsibility.

Contract	Purpose
`ObservationEventV1`	Individual incoming physiological observation
`ObservationWindowV1`	Adaptive gateway evidence packet
`PatientKBRecordV1`	Runtime-safe patient record
`ClinicalContextV1`	Patient Context Expert output

Contract	Purpose
`KnowledgeRuleV1`	Versioned clinical or physiological knowledge
`BaselineStateV1`	Personalized baseline state
`ClinicalScoreV1`	Partial NEWS2 scoring
`ExpertInputV1`	Physiological expert input
`ExpertEvidenceV1`	Structured physiological expert output
`PhysiologicalManifestV1`	Textual virtual physiological representation
`EncoderFrameV1`	Ordered numerical features
`ODEParameterStateV1`	Constrained physiological model parameters
`PatientTwinStateV1`	Observed, estimated and predicted states
`NeuralForecastV1`	Forecast and uncertainty
`TrustAssessmentV1`	Evidence reliability and robustness
`SafetyAssessmentV1`	Independent clinical safety
`ClinicalDecisionV1`	Evidence-constrained recommendation
`ClinicalReportV1`	Structured clinician report
`CarePlanV1`	Approved recovery plan

The previously drafted contracts need to be updated to incorporate the final textual manifest, separate Patient KB context and PINN-ODE state.

---

19. Final division of work

Workstream	Responsibilities
MATLAB & Data Team	Virtual wearable, BIDMC replay, physiological scenarios, batch generation, offline ground truth
Clinical Intelligence Team	Patient KB, Patient Context Expert, Shared Clinical KB, four physiological experts, scoring and baseline rules
AI Architecture Team	Manifest, encoder, PINN-ODE, state estimation, uncertainty, counterfactuals and decision intelligence
Integration & Dashboard Team	Secure gateway, FastAPI, HAPI FHIR, shared interfaces, clinician dashboard, patient dashboard, notifications and evaluation

The MATLAB team’s existing simulator remains usable. The clinical reasoning team’s existing registry and physiological expert work also remain usable.

The main architectural update is the separation of patient context from the textual manifest and the implementation of the PINN-ODE core.

---

20. Final implementation priorities

P0 — Mandatory working prototype

Complete the required problem statement before expanding the research features.

☐ Patient KB with authenticated Patient ID lookup
☐ Patient Context Expert and runtime-safe context output
☐ MATLAB simulator and BIDMC replay
☐ FHIR ingestion and HAPI FHIR storage
☐ Four physiological experts and clinical knowledge
☐ Partial NEWS2 and personalized baseline
☐ Artifact and missing-data handling
☐ Independent clinical safety supervisor
☐ Clinician dashboard and explained alerts
☐ Comparative clinical performance report

P1 — Core innovation

☐ Textual physiological manifest
☐ 264-dimensional context-adaptive encoder
☐ Hybrid PINN-ODE five-minute forecasting
☐ Uncertainty and trustworthy assessment
☐ Evidence-constrained decision system
☐ Authorized clinician notifications
☐ Patient recovery assistant

P2 — Advanced intelligence

☐ Manifest counterfactual branching
☐ 15-minute and 30-minute forecasts
☐ Advanced physiological parameter adaptation
☐ Medication review and approval workflow
☐ Longitudinal recovery intelligence
☐ Comprehensive security and performance testing

Implementation checklist · **0/23**

Use this checklist to track verified completion, not simply documented design.

---

21. Final architecture identity and responsibilities

The complete architecture can be summarized through the responsibility of each intelligence component.

Component	Responsibility
Patient KB	Stores what is known about the patient
Shared Clinical KB	Stores general clinical, physiological and tacit knowledge
Patient Context Expert	Understands who the patient is and what they have undergone
Four Physiological Experts	Understand the patient's current physiological behavior
Textual Manifest	Represents the patient's virtual physiological state
Context-Adaptive Encoder	Converts state and relevant context into numerical features
PINN-ODE	Models how the physiological state evolves
Trustworthy Intelligence	Verifies evidence, uncertainty, trust and robustness
Decision System	Determines the appropriate clinical recommendation
Safety Supervisor	Enforces independent clinical safety requirements
Medication Expert	Supports clinician-controlled intervention
Recovery Expert	Supports the patient's approved recovery journey
Clinician	Reviews and authorizes clinical actions
HAPI FHIR	Preserves standardized clinical records

Final architecture principle

Patient ID retrieves the patient.

The Patient Context Expert understands the patient using the Patient KB and Shared Clinical KB.

The four physiological experts understand the patient's physiological behavior using their skills and applicable tacit knowledge.

The textual manifest reconstructs the virtual physiological state without duplicating the patient's medical history.

The PINN-ODE evolves that state and forecasts possible physiological changes.

The Trustworthy Intelligence layer verifies the resulting intelligence.

The Decision System generates evidence-constrained clinical recommendations.

The Medication and Recovery Experts support human-supervised care.

Architecture freeze

This is the final **V4.1 architecture**, incorporating all the latest decisions:

- Patient ID-based Patient KB retrieval.
- Separate Patient KB and Shared Clinical KB.
- Patient Context Expert as the context controller.
- Four context-adaptive physiological experts.
- Textual-only physiological manifest, without duplicated medical history.
- Manifest combined with hybrid PINN-ODEs.
- Separate context and manifest inputs to the encoder.
- Trustworthy intelligence before the final decision.
- Independent safety path.
- Medication and recovery experts.
- Clinician and patient dashboards.

The architecture is finalized at the system-design level. The remaining implementation dependencies are the exact physiological ODE equations, parameter bounds, updated shared contracts, clinical decision policies and validated model performance.

We should now stop changing the architecture and concentrate on implementing and integrating the working prototype.

If you want, I can:

- Describe next steps for Layer 4 physiological experts
- Outline processing flow from Layer 5 to Layer 7
- Summarize End-to-End patient data flow and usage